

Software Requirements Specification and Other Documents FlatMatch

Group 2

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Table of contents

1. Introduction	4
1.1 Purpose	4
1.2 Document Conventions	4
1.3 Intended Audience and Reading Suggestions	5
1.4 Product Scope	5
1.5 References	6
2. Overall Description	6
2.1 Product Perspective	6
2.2 Product Functions	7
2.3 User Classes and Characteristics	7
2.4 Operating Environment	8
2.5 Design and Implementation Constraints	8
2.6 User Documentation	9
2.7 Assumptions and Dependencies	9
3. External Interface Requirements	9
3.1 User Interfaces	9
3.2 Hardware Interfaces	10
3.3 Software Interfaces	10
3.4 Communications Interfaces	10
3.5 User Interface Mock Up	11
4. Functional Requirements	11
4.1 User Registration, Login & Logout	11
4.2 User Profile	12
4.3 Homepage	12
4.4 User HDB Flats Bookmarks	12
4.5 HDB Listings Overview	12
4.6 HDB Listing detailed Information	13
4.7 HDB Regular Filtering	13
4.8 HDB Amenities Display	13
4.9 HDB Amenities Filtering	13
4.10 Affordability Rating	13
4.11 Data Provenance	14
5. Other Nonfunctional Requirements	15
5.1 Usability	15
5.2 Performance	15
5.3 Reliability	15
5.4 Security	15
5.5 Accessibility	15
5.6 Maintainability	15

5.7 Traceability	15
Appendix A: Glossary	16
Data Dictionary	16
Affordability Rating	16
Appendix B: Sequence Diagram	18
Appendix C0: Use Case Diagram	27
Appendix C1: Use Case Descriptions	28
UC.01 Login	28
UC.02 Register	30
UC.03 View User Information	32
UC.04 Edit User Information	35
UC.05 Change Password	37
UC.06 View HDB Listings	39
UC.07 Search by Neighbourhood	42
UC.08 Filter	44
UC.09 Manage Bookmarks	46
UC.10 View Unit Details	49
UC.11 View Local Amenities	50
UC.12 View Affordability	52
UC.13 Use HDB Finder	54
Appendix D0: Architecture Diagram	56
Appendix D1: UML Class Diagram	57
Appendix E: Test Case	59
Black Box Testing Approach	59
Test Case 1: User register account and log in with valid and invalid credentials	59
Test Case 2: User information, change password and log out	60
Test Case 3: Homepage	61
Test Case 4: Bookmark Listings	62
Test Case 5: Featured Listings and Listings Overview	62
Test Case 6: HDB Listing detailed information with HDB Amenities Display	63
Test Case 7: View all Flats filtering	64
Test Case 8: HDBFinder filtering	65
White Box Testing – Basis Path Testing	66

1. Introduction

1.1 Purpose

We aim to develop an HDB selection website that enables Singapore residents to evaluate both resale and BTO flats. The website will leverage official open datasets to provide accurate and up-to-date information for each HDB listing.

Users will be able to input their personal preferences (e.g., budget, location, number of rooms) to receive tailored recommendations on suitable flats. The system's goal is to allow users to comfortably search, compare, and assess flats based on price, location, nearby amenities, and overall affordability.

1.2 Document Conventions

1.2.1 The SRS is written in clear, concise language with consistent terminology throughout. Technical terms and acronyms are defined in the Glossary (Appendix A) to ensure clarity.

1.2.2 Section numbering follows a hierarchical structure (e.g., 1, 1.1, 1.2, etc.) so that high-level requirements are easily traceable to detailed specifications. Higher-level requirement priorities are inherited by their detailed sub-requirements, unless explicitly specified otherwise.

1.2.3 Important terms, file names, or API names are highlighted using italic or bold text where appropriate. All such formatting is used consistently throughout the document.

1.2.4 Requirement statements are formatted as complete sentences and provided with unique identifiers or numbers to support traceability and facilitate review, testing, and change management.

1.2.5 Diagrams (e.g., use case and sequence diagrams) and test cases are included in the Appendices to provide additional context and modelling support, adhering to industry-standard notations.

1.3 Intended Audience and Reading Suggestions

Developers: To understand the functional and technical specifications necessary for implementing the HDB Finder system.

Testers: To design and execute test cases based on the specified system behaviours and requirements.

End Users: To gain an overview of the system's features, purpose, and expected user experience when searching for suitable flats.

1.4 Product Scope

The HDB Finder website will serve as an interactive platform that simplifies the process of identifying suitable HDB flats for potential homeowners. It will integrate data from official public sources to ensure reliable and accurate listings, covering both resale and BTO flats. The scope of this project focuses on developing a responsive, user-friendly, and data-driven web application tailored to the needs of Singapore residents seeking affordable and well-located housing options.

The system will feature key functionalities:

1. Personalized flat recommendations based on User's particulars
2. Search and filtering of HDB Flats by criteria (price range, location, flat type, amenities)
3. Save a listing for viewing later
4. Viewing of nearby amenities of the listed HDB Flat

1.5 References

1. Singapore Housing & Development Board, "*Resale Flat Prices Based on Registration Date (from Jan 2017 Onwards)*", data.gov.sg, 2024. <https://data.gov.sg/dataset/resale-flat-prices>
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2. Overall Description

2.1 Product Perspective

The HDB Finder website is a self-contained new product that allows the users to evaluate HDB flats. It is neither a component of a larger system, nor a replacement for an existing system.

2.2 Product Functions

The following are the major functions of the HDB Finder website. A more detailed discussion can be found in Section 4.

1. The website allows the user to register for an account and browse as a logged-in user. Alternatively, the website allows the user to browse as a guest.
2. The website allows the user to input their personal information, such as household income and budget.
3. The website displays HDB listings.
4. The website allows the user to search and filter the HDB Listings.
5. The website allows the user to bookmark desirable HDB units.
6. The website displays information about HDB units, such as the neighbourhood and price.
7. The website evaluates the affordability of HDB units for the user, based on information acquired in Function 2.
8. The website recommends the best units for the users based on the user's preference.

2.3 User Classes and Characteristics

User Classes	Purpose	Frequency of Use	Subset of Product Functions Used by this User Class
HDB Buyers	Evaluate HDB flats for purchase	High (weekly)	Registration, Affordability, Display Listings, Search & Filter, Bookmark, Display Unit Details
HDB Sellers	Monitor the market price of the flats they own	Moderate (monthly)	Registration, Display Listings, Search & Filter, Bookmark, Display Unit Details

Property Investors	Evaluate HDB flats for investment	Moderate (monthly)	Registration, Display Listings, Search & Filter, Bookmark, Display Unit Details
Researchers & Policy Makers	Evaluate the housing market and monitor the impact of policies	Moderate (quarterly)	Registration, Display Listings, Search & Filter, Bookmark, Display Unit Details

Our most important user class is HDB Buyer. Therefore, we provide functionalities that cater to the needs of individual buyers, such as the Affordability functionality, which evaluates whether a unit is affordable for an individual buyer, based on the buyer's income and budget. Other user classes may have needs for different functionalities. For example, researchers may focus less on the affordability of a particular flat, but more on understanding the general market price trend. However, since our most important user class is HDB Buyer, we prioritise their needs over other user classes.

2.4 Operating Environment

This software will operate on all hardware platforms (Windows, Linux and MacOS).

This software will operate on all browsers (Chrome, Firefox etc).

2.5 Design and Implementation Constraints

1. The website needs to comply with the Personal Data Protection Act (PDPA) of Singapore. It needs to implement secure data handling, encryptions, explicit consent forms and privacy statements.
2. API prices need to be considered, because the website uses external APIs to provide map visualisation.
3. There are no programming language constraints. However, whichever languages or technological stacks are chosen, good software engineering practices need to be observed.
4. The primary display language of the website should be English. Other languages, such as Chinese, Malay and Tamil, should also be made available for greater accessibility.

2.6 User Documentation

A demo or promotional video will be created to introduce users to the application and demonstrate its key features and usage. This video will serve as a simple, visual guide to help users understand how to navigate and interact with the website. The interface is intentionally designed to be straightforward and intuitive, with minimal use of symbols, icons, or hidden menus. This ensures that even non-tech-savvy users can easily explore the platform, understand each function, and perform desired actions without confusion.

2.7 Assumptions and Dependencies

It is assumed that all BTO and resale flat data provided by official government sources are accurate and up to date, ensuring that users receive reliable information.

It is assumed that the listed BTO projects remain available for viewing at the time users access the website, although this may not always reflect real-time availability.

3. External Interface Requirements

3.1 User Interfaces

The user starts on a login page which allows the user to login to their already registered account or create a new account. The login page requires the user to input their username and their password. On the register page, the user creates an account with a unique username, password, and a confirm password to ensure that the user entered the correct password. This allows users to save their user information, preferences and bookmarked flats for future visits.

Once logged in, users can access a personal information page, where they can input their monthly income, downpayment budget, and other relevant details. This information enables the system to evaluate the affordability of various flats and recommend suitable options based on the user's financial capacity.

In the listings page, users can browse both BTO and resale flats. Users are able to apply discrete filters such as location, flat type (e.g., 3-room, 4-room, 5-room) and priority filters such as affordability (based on user information) and proximity to amenities (public transport, schools, etc.). If users see a listing they are interested in, they are able to click on the listing to view the full details of the flat. Users can also bookmark their preferred flats, which will appear on a dedicated *Bookmarks* page.

The interface is designed to be simple, responsive, and user-friendly, working smoothly on both desktop and mobile devices. It features clear navigation, labelled buttons, and accessible layouts to ensure that users of all experience levels can comfortably explore and interact with the platform.

3.2 Hardware Interfaces

The system is fully web-based and designed to operate on standard computing devices without the need for any specialized hardware. Users can access the website through any modern desktop, laptop, tablet, or mobile device equipped with an internet connection. The interface is optimized for both mouse and touchscreen inputs, ensuring usability across Windows, macOS, Android, and iOS platforms. On the server side, the application is hosted on a cloud environment that supports Node.js and MongoDB, requiring minimal hardware resources. The database is managed through a cloud-based MongoDB server, which provides high availability and scalability without physical hardware management on the user's end.

3.3 Software Interfaces

The system integrates multiple software components that enable seamless communication between the front-end, back-end, and database layers. The front-end is developed using React.js, which manages the user interface and interacts with the server through RESTful APIs. These APIs are handled by the Node.js and Express.js backend, which processes requests such as login authentication, flat retrieval, and affordability calculations. The backend connects to the MongoDB database using Mongoose, an Object Data Modeling (ODM) library that defines schemas for user data, listings, and bookmarks. Additionally, the website uses Next.js for server-side rendering and routing, improving performance and SEO. External data sources, such as HDB Open Data and Data.gov.sg APIs, are also integrated to ensure that flat listings and pricing data are accurate and up to date. Configuration files such as `.env.local` store sensitive environment variables securely, while TypeScript ensures strong type safety throughout the application, reducing runtime errors and improving code maintainability.

3.4 Communications Interfaces

The system communicates using standard HTTPS protocols to ensure secure and reliable data exchange between the client, server, and external sources. The frontend (React.js) interacts with the backend (Node.js and Express.js) through RESTful APIs, transmitting data in JSON format. When users log in, register, or update personal information such as income and budget, the frontend sends requests to API routes, which the backend processes and stores in MongoDB via Mongoose. Flat listings and related details are retrieved dynamically from external data sources such as HDB Open Data and Data.gov.sg APIs, rather than being stored locally. This approach ensures that users always access the most accurate and up-to-date housing information while keeping database storage lightweight and focused on user data.

3.5 User Interface Mock Up

The user interface will be designed to provide a simple, intuitive, and user-friendly experience. It will maintain a consistent color scheme, clear typography, and well-organized navigation menus. The interface will be fully responsive and accessible on both desktop and mobile devices. The mock-up will serve as a foundational design reference, guiding the development of the software to ensure that the final product closely resembles the proposed layout and functionality.



4. Functional Requirements

4.1 User Registration, Login & Logout

4.1.1 The system shall allow a user to register an account with a unique username and password.

4.1.2 The system shall validate that all required fields are completed before creating the account.

4.1.3 The system shall allow a registered user to log in using a valid username and password, validating credentials before granting access.

4.1.4 The system shall allow a logged-in user to log out.

4.2 User Profile

4.2.1 The system will allow the user to update their user information with the following attributes: yearly household income, and age.

4.2.2 The system will allow the user to update their user preference with the following attributes: flat type, preferred area and down payment budget.

4.2.3 The system shall validate that all required fields are completed before saving the profile.

4.3 Homepage

4.3.1 The homepage shall display featured neighbourhoods based on the user's preferences.

4.3.2 The homepage shall provide quick access to search and filter HDB listings using both standard and amenities-based filters.

4.3.3 The homepage shall provide quick access to the user's Bookmarked Listings.

4.3.4 The homepage shall provide quick access to user account functions, including login and logout.

4.4 User HDB Flats Bookmarks

4.4.1 The system shall allow users to access the bookmark function, ensuring that only logged-in users can use it.

4.4.2 Logged-in users shall be able to save desirable listings to their Bookmarked Listings for later viewing.

4.4.3 Logged-in users shall be able to remove listings from their Bookmarked Listings.

4.4.4 Logged-in users shall be able to view all listings in their Bookmarked Listings.

4.4.5 Logged-in users shall be able to view detailed information for any listing saved in their Bookmarked Listings.

4.5 HDB Listings Overview

4.5.1 The homepage shall display featured neighbourhoods based on the user's preferences.

4.5.2 The listings shown to the user shall include the following attributes: neighbourhood, flat type, cost price, and affordability rating.

4.5.3 The user shall be able to select any listing to view detailed information about that HDB flat.

4.6 HDB Listing detailed Information

4.6.1 The system shall display the following attributes for each flat listing: neighbourhood, flat type, cost price, block, street, storey, floor area, model, lease commencement date, remaining lease, and listing posted month.

4.6.2 The system shall display an affordability rating for each listing based on the user's information and preferences.

4.6.3 The system shall display nearby amenities for each listing, using HDB Amenities Display.

4.7 HDB Regular Filtering

4.7.1 The system shall allow the user to filter HDB listings by the following attributes, which may be used individually or in combination: neighbourhood, price range, and flat type.

4.8 HDB Amenities Display

4.8.1 The HDB Amenities Display shall include the following filtering attributes: transport, schools, clinics/polyclinics, supermarkets, hawker centres, parks, libraries, and community clubs.

4.8.2 The system shall allow the user to filter and view amenities within a 1 km radius of the selected flat.

4.8.3 The system shall allow the user to select any amenity marker to calculate and display walking distance and estimated walking time.

4.8.4 The system shall allow the user to explore the surrounding area using click-and-drag navigation.

4.9 HDB Amenities Filtering

4.9.1 The system shall allow the user to assign a priority rating (on a scale of 1 to 10) to the following attributes: distance to MRT, distance to schools, distance to medical facilities, and affordability rating.

4.9.2 The system shall allow the user to select up to three towns and specify the desired flat type for filtering.

4.10 Affordability Rating

4.10.1 The system shall calculate the down payment required for each flat based on the total flat price which is 25% of the total cost of the house according to HDB's official housing loan.

4.10.2 The system shall evaluate whether the user can afford the required down payment and monthly repayment, displaying "✓ Yes" or "✗ No" accordingly.

4.10.3 The system shall compute monthly repayment using the selected tenure (in years) and loan interest rate according to HDB's loan repayment period and applicable interest rate.

4.10.4 The system shall calculate the 30% income limit (30% of the user's monthly income) and ensure the monthly repayment does not exceed this threshold.

4.10.5 The system shall generate an Affordability Score on a scale of 1 to 10, based on both the user's ability to pay the down payment and the monthly repayment ratio to income.

4.10.6 The system shall display an overall Affordability Status (e.g., Affordable, Moderate, Unaffordable) based on the computed score.

4.11 Data Provenance

4.11.1 The system shall display the dataset source and last update timestamp for each information type.

5. Other Nonfunctional Requirements

5.1 Usability

5.1.1 At least 80% of first-time users must be able to perform a search within 2 minutes of first use.

5.2 Performance

5.2.1 Search queries shall return results in ≤ 1.5 seconds (95th percentile).

5.2.2 Map markers shall render within 2.5 seconds (95th percentile).

5.3 Reliability

5.3.1 If the live carpark API fails, the system shall fall back to static carpark data.

5.4 Security

5.4.1 User profile data shall be encrypted at rest and transmitted via TLS.

5.5 Accessibility

5.5.1 The interface shall comply with WCAG 2.1 AA standards.

5.6 Maintainability

5.6.1 All external data sources shall be refreshable via scheduled jobs with monitoring.

5.7 Traceability

5.7.1 Each requirement shall have a unique identifier that can be linked to design elements, test cases, and code modules.

Appendix A: Glossary

Data Dictionary

Term	Definition
User	An individual accessing the website to search for HDB units.
Logged-in User	A User who is currently logged in.
Guest	A User who is currently not logged in.
User Profile	Personal details provided by the user, such as name, email, monthly income, household size, and downpayment budget. This information is used to calculate housing affordability and personalize recommendations.
Resale Transaction	Refers to a record of an HDB resale flat transaction obtained from official datasets (e.g., Data.gov.sg).
Block	Represents a physical, localised cluster of HDB units.
Carpark	Refers to nearby carpark associated with an HDB block.
MRT Exit	Represents a specific exit point of an MRT station.
School	Refers to nearby primary, secondary, or tertiary educational institutions.
Affordability Rating	A computed score based on the user's monthly income, budget, and resale flat price. Indicates how financially suitable a given flat is for the user.
Bookmark	A record of flats saved by the user for later viewing. Each bookmark links the user's account to a specific flat listing
Provenance	Refers to the source and verification details of a dataset (e.g., HDB Open Data, Data.gov.sg).
Unit	Represents an individual flat unit within a HDB block.

Affordability Rating

The Affordability Rating represents how affordable a flat is for an applicant, based on the ratio between the applicant's monthly income and the required monthly loan repayment.

Repayment Period:

The repayment period is capped at the shortest of the following:

- 25 years;
- 65 years minus the applicants' average age; or
- Remaining lease of the flat minus 20 years.

Applicable Interest Rate:

HDB uses the higher of:

- The interest rate floor (currently 3.0% per annum); or
- The prevailing HDB housing loan interest rate.
- The concessionary interest rate is pegged at 0.1%-point above the prevailing CPF Ordinary Account interest rate and reviewed quarterly.

Monthly Instalments:

Monthly instalments are capped at up to 30% of the applicants' monthly income.

The affordability rating is based on the applicant's ability to afford the required monthly repayment, expressed as a percentage of monthly income compared to the required monthly payment.

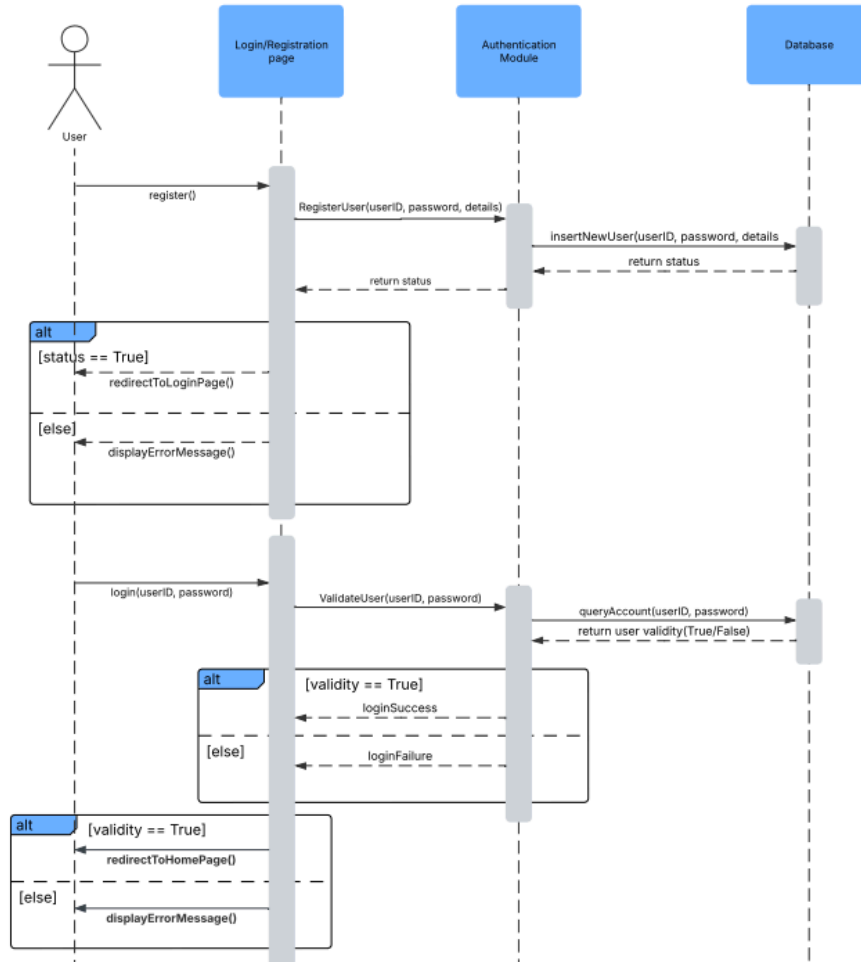
- If the applicant can afford the monthly repayment fully or with a surplus ($\geq 100\%$), a higher score is assigned.
- If the applicant cannot afford the repayment ($< 100\%$), a lower score is assigned.

Affordability Level	Condition	Affordability Rating
	200% or more of required amount	10/10
	170% or more	9/10
	140% or more	8/10

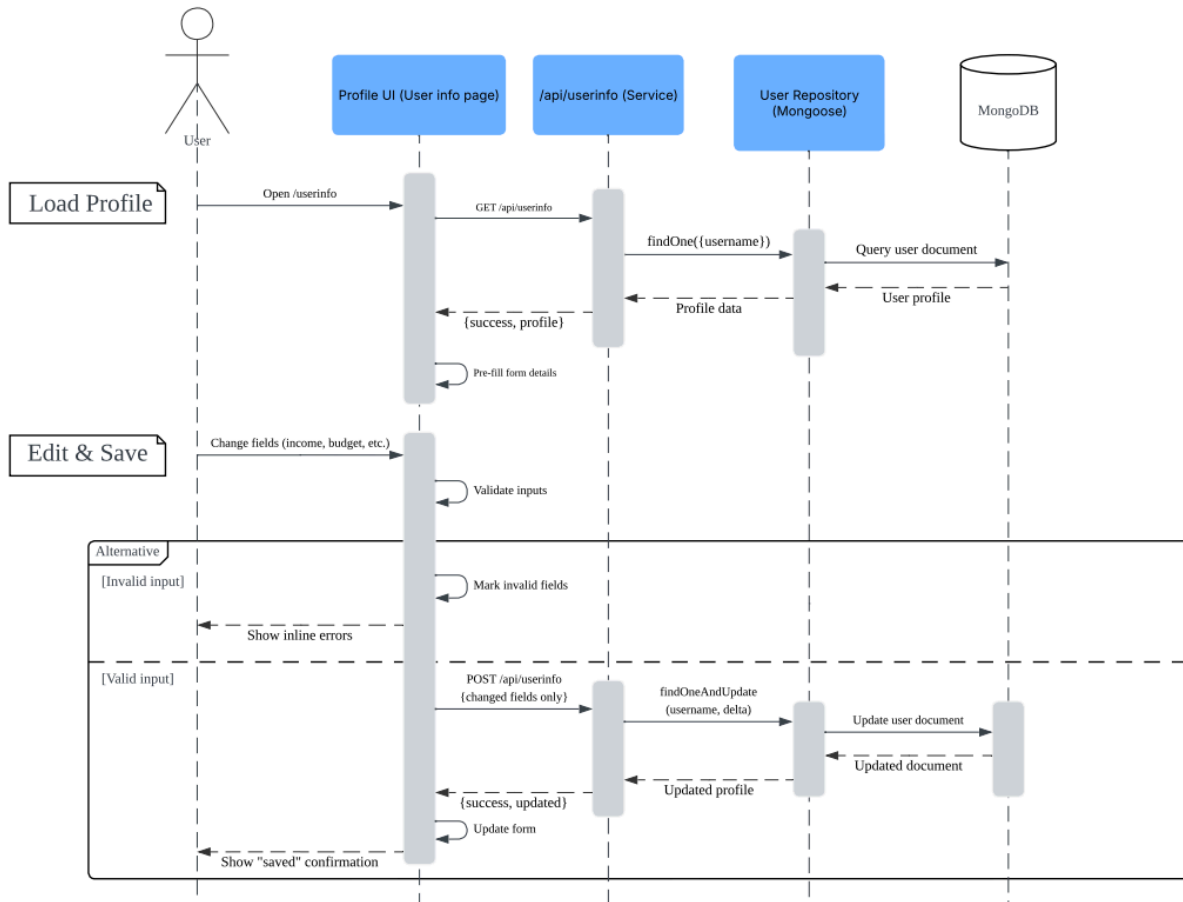
Can afford ($\geq 100\%$)	120% or more	7/10
	105% or more	6/10
	Exactly 100 %	5/10
Cannot afford ($< 100\%$)	95% or more	4/10
	75% or more	3/10
	70 % or more	2/10
	Below 70%	1/10

Appendix B: Sequence Diagram

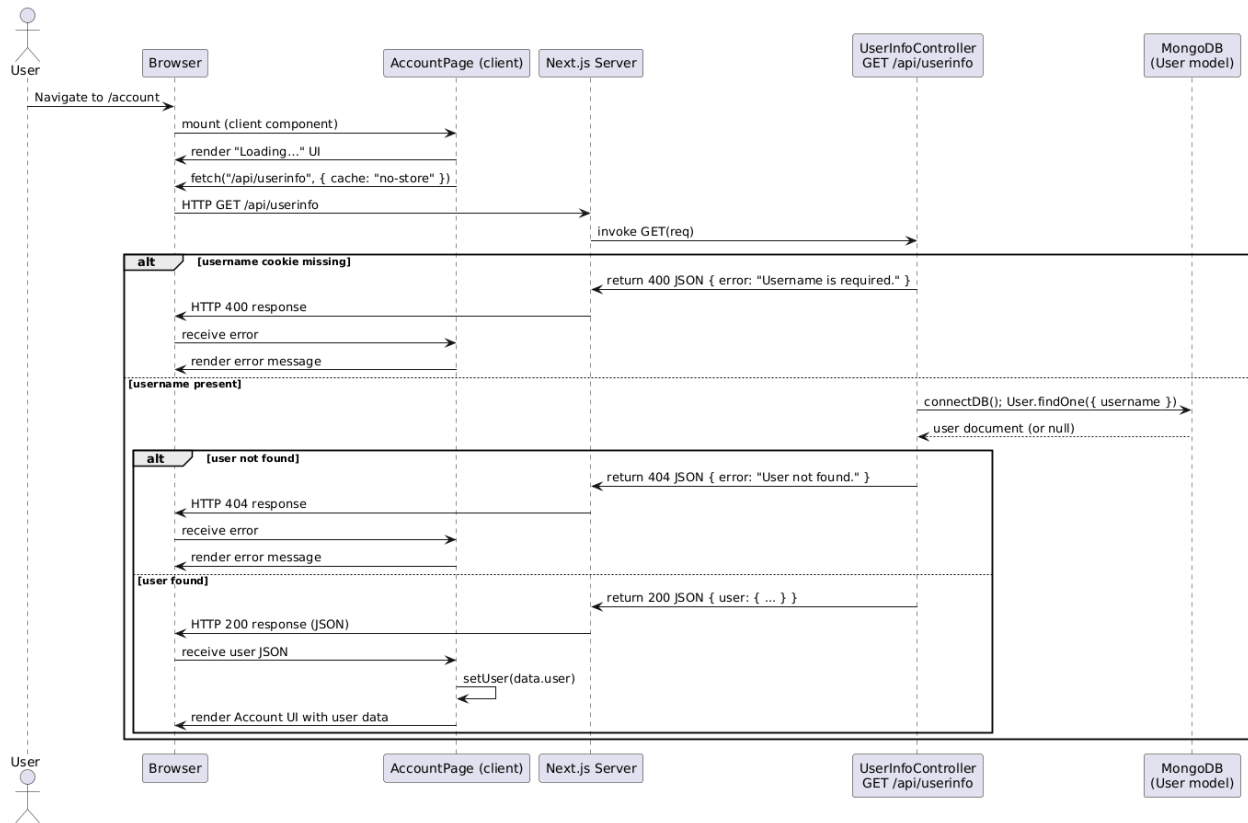
Registration and Login



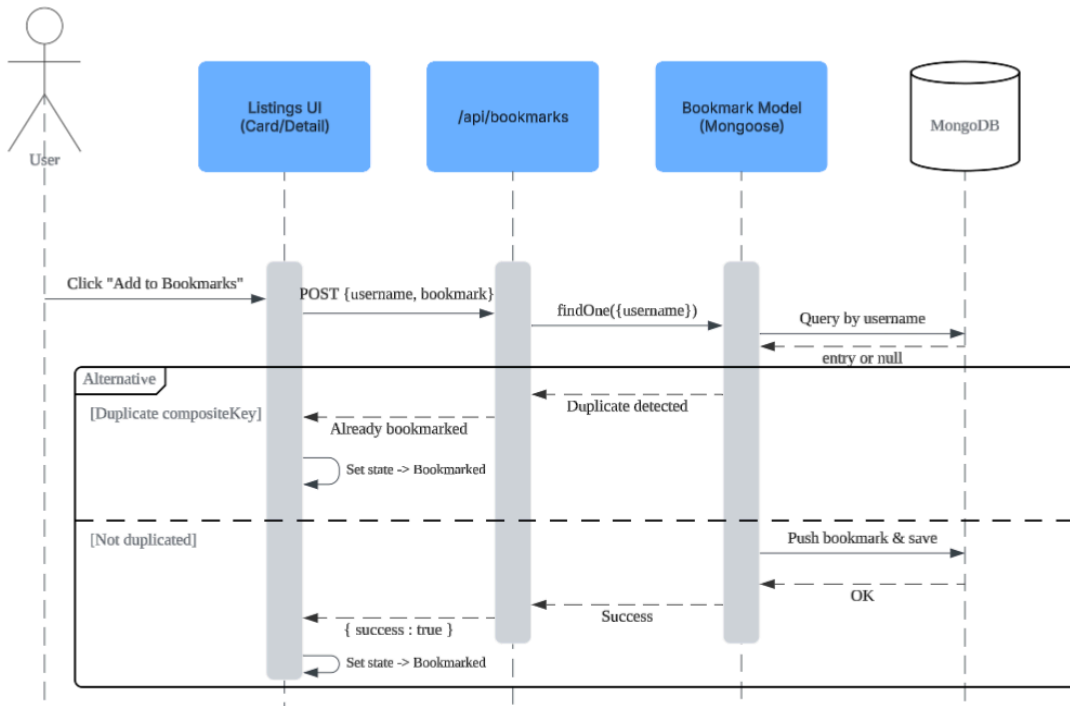
Manage User Information



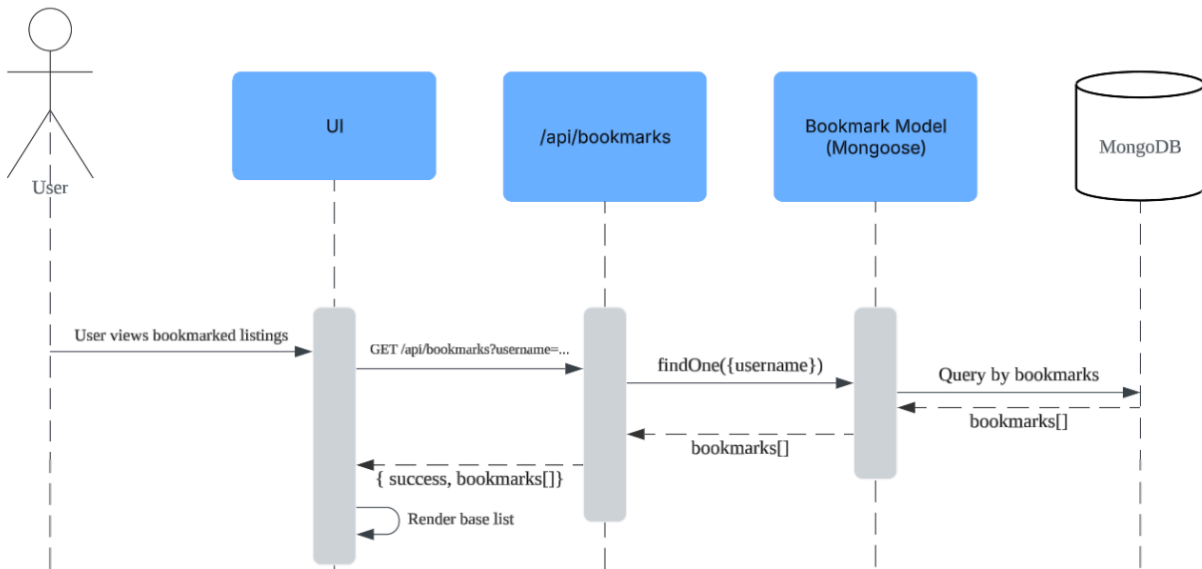
View Account



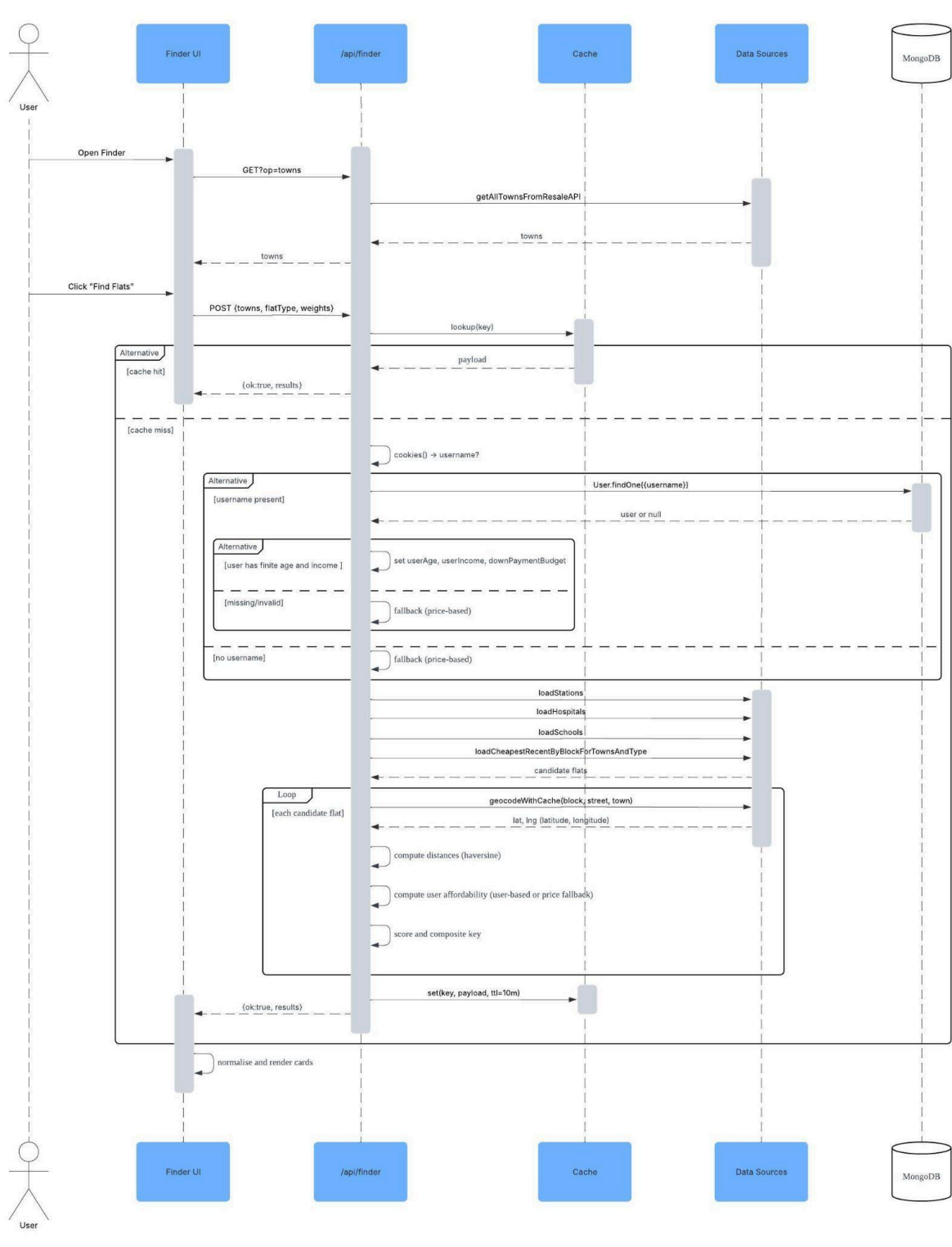
User Bookmarks a HDB Listing



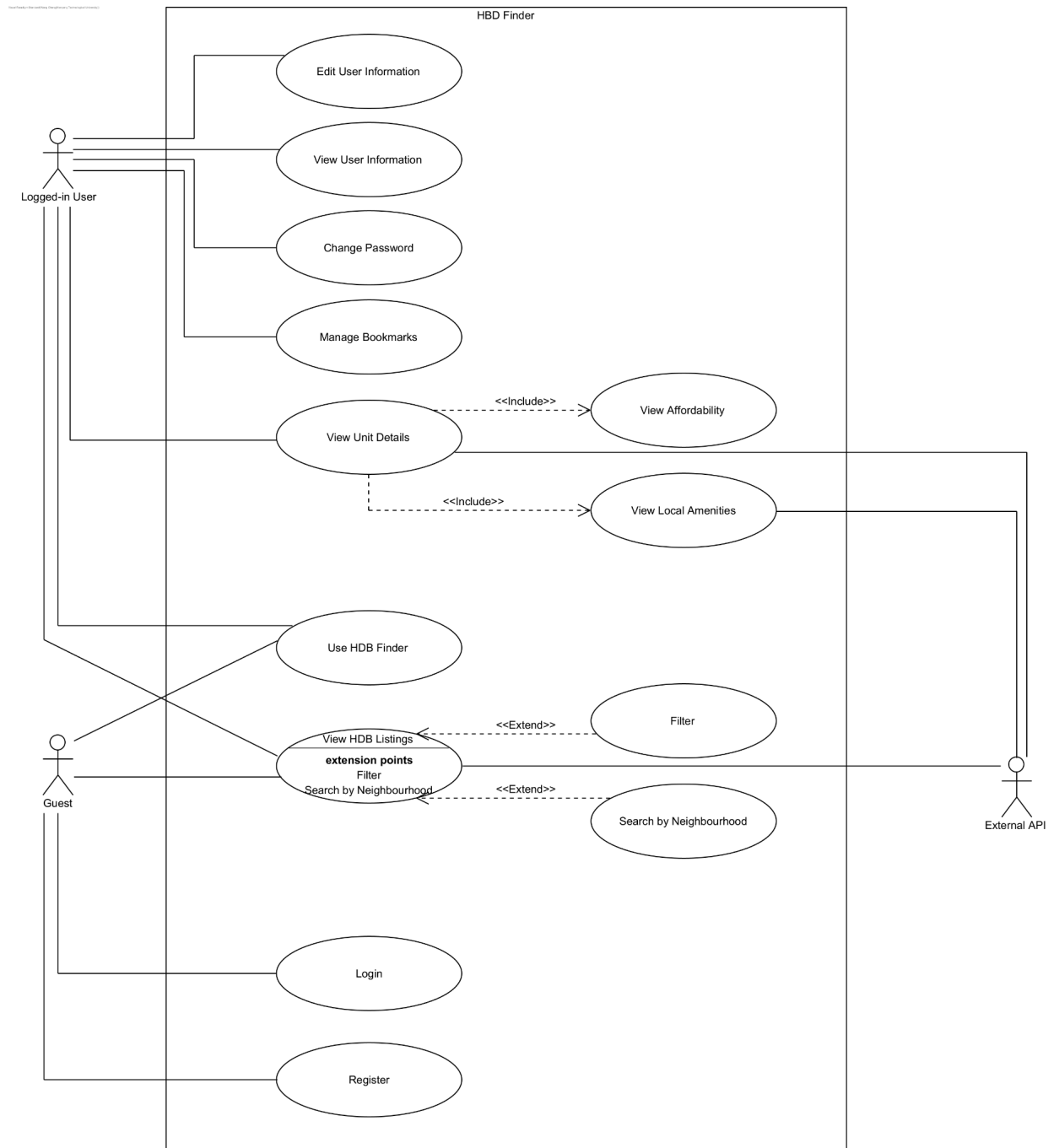
User Views Bookmarks



HDB Finder



Appendix C0: Use Case Diagram



Appendix C1: Use Case Descriptions

UC.01 Login

Use Case ID:	UC.01		
Use Case Name:	Login		
Created by:	Wang Cheng	Last Updated by:	Wang Cheng
Date Created:	25/08/2025	Date Last Updated:	09/11/2025
Actor:	Guest		
Description:	Allow a user to log in to their account.		
Preconditions:	1. The User has an existing account. 2. The User is currently not logged in.		
Postconditions:	1. The User is successfully logged in. 2. The top-right corner displays the username.		
Priority:	High		
Frequency of Use:	Low – Estimated 1 time per user per hour		

Flow of Events:	1. The Guest clicks on the “Login” button on the root page. 2. The System brings the Guest to the Login Page, where two blanks are shown - one for username and one for password. There is also a Register (UC.02) link below the blanks. 3. The Guest enters their username and the password 4. The Guest clicks on the “Login” button 5. The System logs the Guest in 6. The User is now a logged-in user, and they are redirected to the home page.
Alternative Flows:	UC.01.AF-S2: If the Guest does not have an account - The Guest clicks on the “Register” below the blanks - The Guest is redirected to the Register Page (UC.02)
Exceptions:	UC.01.EX.01: If username or password is missing - The System shows a “Please fill out this field.” reminder below the empty field. UC.01.EX.02: If the credentials are wrong - The System shows “Login failed. Please try again.”
Includes:	NIL
Special Requirements:	1. Visual feedback on submit within 100ms (button disabled, replaced with “Logging in…”). 2. Typical login round-trip ≤ 5 s; worst-case user-facing timeout message at 10 s. 3. On success, top bar updates within 300ms of navigation to Home.

Assumptions:	NIL
Notes and Issues:	NIL

UC.02 Register

Use Case ID:	UC.02		
Use Case Name:	Register		
Created by:	Wang Cheng	Last Updated by:	Wang Cheng
Date Created:	25/08/2025	Date Last Updated:	09/11/2025
Actor:	Guest		
Description:	Allows the User to create an account with a unique username and their password.		
Preconditions:	1. The Guest is currently not logged in		
Postconditions:	1. A new user account is created and stored. 2. The user is redirected to the Login Page.		

Priority:	High
Frequency of Use:	Low – 1 time per user
Flow of Events:	1. The Guest clicks on the “Register” button on the root page 2. The System redirects the User to the Registration Page, with three blank fields: Username; Password; and Re-enter Password. There is also a Login link below these fields. 3. The Guest enters username, password, and their password once more for confirmation. 4. The Guest clicks the “Register” button. 5. The System creates the account and redirects the Guest to the Login Page (UC.01)
Alternative Flows:	UC.02.AF-S2: If the Guest already has an account - The Guest clicks on the “Login” link below the blanks - The System redirects the Guest to the Login Page (UC.01).
Exceptions:	UC.02.EX.01: if the username is already taken - The System displays “username already taken” UC.02.EX.02: if the password is different from the re-entered password - The System displays “Passwords do not match” UC.02.EX.03: if the password is less than 5-characters long

	- The System displays “Please lengthen this text to 5 characters or more (you are currently using x characters)”, with ‘x’ representing the current number of characters.
Includes:	NIL
Special Requirements:	1. Field validation and inline messages within 150ms of blur. 2. Submission completes within 2.0 s in typical conditions. 3. Passwords stored securely (hashed); no password shown by default (toggle eye icon).
Assumptions:	The Guest does not have an existing account
Notes and Issues:	NIL

UC.03 View User Information

Use Case ID:	UC.03		
Use Case Name:	View User Information		
Created by:	Kenny Lim	Last Updated by:	Wang Cheng
Date Created:	4/10/2025	Date Last Updated:	09/11/2025
Actor:	Logged-in User		

Description:	Show the User's stored profile and preferences on the Account Page
Preconditions:	1. The User is logged in.
Postconditions:	1. The Account Page displays the User's information, including username, income and citizenship
Priority:	High
Frequency of Use:	Medium – Estimated 2 to 3 times per user per hour
Flow of Events:	1. The User is on the home page. 2. The User clicks the top-right button, which displays their username. 3. The System displays a drop-down menu. 4. The User clicks "Account". 5. The System redirects the User to the Account Page showing the User's stored profile and preferences.
Alternative Flows:	NIL

Exceptions:	UC.03.EX.01: If the User is not logged in - The System shows a status message “Failed to fetch user info.” UC.03.EX.02: If the account is not found - The System shows a status message “Failed to fetch user info.” UC.03.EX.03: If a network error occurs - The System shows a status message “Failed to fetch user info.”
Includes:	NIL
Special Requirements:	1. Speed: First visible content ≤ 1.0 s 2. Clarity: Each section (Profile / Housing Preferences / Financials) uses bold headings and consistent spacing; values appear in bordered chips or rows. 3. Accessibility: The Account button and dropdown are keyboard reachable with visible focus.
Assumptions:	A signed-in User has at least a username stored; other fields may be empty.
Notes and Issues:	NIL

UC.04 Edit User Information

Use Case ID:	UC.04		
Use Case Name:	Edit User Information		
Created by:	Wang Cheng	Last Updated by:	Wang Cheng
Date Created:	25/08/2025	Date Last Updated:	09/11/2025
Actor:	Logged-in User		
Description:	Allow the User to fill in or edit their information and preferences (income, citizenship, age, flat type, down payment budget, preferred area)		
Preconditions:	1. The User is logged in.		
Postconditions:	1. The information and preferences are successfully stored. 2. The System displays a message “Saved successfully”.		
Priority:	Medium		
Frequency of Use:	Low – Estimated once per user per session		

Flow of Events:	1. The User is on the Account Page. 2. The User clicks the “Edit Info” button beneath the information sections. 3. The System shows the User Information form page with labelled inputs and dropdowns. 4. The System pre-fills fields with the User’s current stored values (if any): - Yearly Household Income - Citizenship - Age - Flat Type - Down Payment Budget - Preferred Area 5. The User reviews the fields and updates one or more values. 6. The User clicks Save. 7. The System validates the inputs (e.g., numeric checks for income/budget). 8. The System stores the values and shows a green success status: “Saved successfully”. 9. The User may return to Account (e.g., via the header navigation) to see the updated values reflected.
Alternative Flows:	UC.04.AF-S1: If the User is on the home page - The User clicks the top-right button, which displays their username. - The System displays a drop-down menu. - The User clicks “User Info” - Go to step 3 of the normal flow.

Exceptions:	· UC.04.EX.01 – Invalid income: The System shows an inline error: “Yearly household income must be an integer.” The offending field is highlighted; no data is saved. · UC.04.EX.02 – Invalid budget: The System shows an inline error: “Budget must be an integer if provided.” The offending field is highlighted; no data is saved. · UC.04.EX.03 – Failed to load current values: A status banner appears at the top: “Failed to fetch user info.” · UC.04.EX.04 – Network/server error on save: A status banner appears: “Failed to save user info. Please try again.” Entered values remain on screen so the User can retry.
Includes:	NIL
Special Requirements:	1. Successful save completes within ≤ 1.0 s on typical broadband; success banner appears without full page reload. 2. Inline error text appears beneath the field within 300ms of the server response; focus moves to the first invalid field.
Assumptions:	NIL
Notes and Issues:	NIL

UC.05 Change Password

Use Case ID:	UC.05		
Use Case Name:	Change Password		
Created by:	Wang Cheng	Last Updated by:	Wang Cheng
Date Created:	25/08/2025	Date Last Updated:	09/11/2025
Actor:	Logged-in User		
Description:	Allow the User to change their password.		
Preconditions:	1. The User is logged in.		
Postconditions:	1. The password is changed successfully 2. The System displays “Password updated successfully”		
Priority:	Medium		
Frequency of Use:	Low – Estimated once per user per session		

Flow of Events:	1. The User is on the Account Page 2. The User clicks “Change Password” 3. The System redirects the User to Change Password Page 4. The User enters the current password, new password and again the new password. 5. The User clicks “Update Password” 6. The System displays “Password updated successfully”
Alternative Flows:	NIL
Exceptions:	UC.05.EX.01: If the new password is the same as the current password - The System displays “The new password cannot be the same as the current password” UC.05.EX.02: If the new password and the re-entered new password are different - The System displays “The passwords do not match”.
Includes:	NIL
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

UC.06 View HDB Listings

Use Case ID:	UC.06		
Use Case Name:	View HDB Listings		
Created by:	Wang Cheng	Last Updated by:	Wang Cheng
Date Created:	25/08/2025	Date Last Updated:	09/11/2025
Actor:	Logged-in User or Guest		
Description:	Displays a growing list of HDB flats in cards as the User scrolls		
Preconditions:	1. The dataset endpoint is reachable. 2. The User can open the HDB Listings screen		
Postconditions:	1. A list of HDB records is visible; more records load automatically near the bottom. 2. Each card shows town, flat type, price and opens its detailed view when clicked (UC.10).		
Priority:	High		
Frequency of Use:	High – Estimated 5 to 10 times per user per hour		

Flow of Events:	1. From the Home page, the User clicks the “View all Flats” button beneath the search bar. 2. The System opens the Listings screen and shows a “Loading...” indicator. 3. The System displays the first batch of HDB cards (town, flat type, price). 4. The User scrolls, looking through the HDB flats 5. When near the bottom, the System shows “Loading...” and more cards are appended. 6. When all available items have been loaded, the System shows “No more listings.”
Alternative Flows:	UC.06.AF-S1: If the User is not on the home page - On any page that shows the header with a menu button, the User taps the menu button. - In the menu, the User selects “View all Flats” (or “View All Listings”). - The System opens the Listings screen and continues with Steps 2–6 of the normal flow. UC.06.AF-S4-1: If the User searches by neighbourhood - Go to UC.07 (Search by Neighbourhood) UC.06.AF-S4-2: If the User applies filters - Go to UC.08 (Filter) UC.06.AF-S4-3: If the User clicks a card to view its details - The System redirects the User to the unit details page - Go to UC.10 (View Unit Details)

Exceptions:	NIL
Includes:	NIL
Special Requirements:	1. First results visible $\leq 1.5s$, first-time users should be able to perform a search within 2 minutes 2. Page should be able to display at least 10 listings at a time to reduce the need for excessive scrolling on the User's end
Assumptions:	1. The data from the dataset where HDB data is fetched from is up to date and accurate
Notes and Issues:	NIL

UC.07 Search by Neighbourhood

Use Case ID:	UC.07		
Use Case Name:	Search by Neighbourhood		
Created by:	Wang Cheng	Last Updated by:	Wang Cheng
Date Created:	25/08/2025	Date Last Updated:	09/11/2025
Actor:	Logged-in User or Guest		
Description:	Allow the User to search by neighbourhood.		
Preconditions:	1. The User is logged in. 2. The User is viewing HDB listings (UC.06).		
Postconditions:	1. The System displays only HDB units from the chosen neighbourhood.		
Priority:	Medium		
Frequency of Use:	High		

Flow of Events:	1. The User is viewing HDB listings. 2. The User enters the name of a neighbourhood in the search bar. 3. The User clicks the “Search” button. 4. The System displays only HDB units from the chosen neighbourhood.
Alternative Flows:	NIL
Exceptions:	NIL
Extends:	UC.06 (View HDB Listings)
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

UC.08 Filter

Use Case ID:	UC.08		
Use Case Name:	Filter		
Created by:	Wang Cheng	Last Updated by:	Wang Cheng
Date Created:	25/08/2025	Date Last Updated:	09/11/2025
Actor:	Logged-in User or Guest		
Description:	Allow the User to filter by price and the number of rooms.		
Preconditions:	HDB Listings page correctly displays the listings (UC.05)		
Postconditions:	The User is able to see the list of HDB units which meet the constraints they have selected via the filters		
Priority:	High		
Frequency of Use:	Medium – Estimated 2 to 5 times per user per hour		

Flow of Events:	1. The User is on the HDB Listings Page. 2. The User enters the price range in the blank. 3. The User selects the number of rooms they want in each HDB unit. 4. The User clicks “Apply Filters” 5. The System displays the list of HDB units which satisfies the filtering conditions.
Alternative Flows:	NIL
Exceptions:	NIL
Extends:	View HDB Listings (UC.06)
Special Requirements:	Results visible $\leq 1.5s$ after filters or sort are applied
Assumptions:	NIL
Notes and Issues:	NIL

UC.09 Manage Bookmarks

Use Case ID:	UC.09		
Use Case Name:	Manage Bookmarks		
Created By:	Kenny Lim	Last Updated By:	Kenny Lim
Date Created:	22/08/2025	Date Last Updated:	19/10/2025

Actor:	Logged-in User
Description:	The Logged-in User adds a unit to Bookmarks from the Unit Details view and can view/remove bookmarks from the Bookmarks screen.
Preconditions:	1. The User is Logged in. . 2. The User can open a Unit Details card (UC.10) and the Bookmarks screen.
Postconditions:	1. User is able to add units to a recorded shortlist 2. User is able to remove added units from the shortlist 3. User is able to shortcut to the detailed view of the flat's attributes (UC.10)
Priority:	Medium
Frequency of Use:	Low – Estimated 1 to 2 times per user per hour

Flow of Events:	1. The User views a Unit Details card. 2. The User clicks Add to Bookmarks. 3. The System stores the bookmark and swaps the button label to Bookmarked. 4. The User opens Bookmarks (a page listing bookmarked units). 5. The System shows bookmarked items in cards (street/flat type/price). 6. The User clicks on one of the cards 7. The System navigates the User to the Detailed View Page (UC.10) 8. The User navigates back to the Bookmark screen, and clicks Remove on an item 9. The System removes the item from the Bookmarks, the list updates immediately
Alternative Flows:	NIL

Exceptions:	1. UC.08.EX.01 – Not signed in, the System cannot fetch bookmarks and shows a red message on the Bookmarks page: “Not logged in.” 2. UC.08.EX.02 – Failed to add bookmark (e.g., duplicate or network). On the unit details page, a red message appears above the card content: “Failed to add bookmark.” The button remains “Add to Bookmarks.” 3. UC.08.EX.03 – Failed to remove bookmark (network/server). On the Bookmarks page, the System shows a browser alert: “Failed to remove bookmark.” The card remains in the list. 4. UC.08.EX.04 – Failed to fetch bookmarks. On the Bookmarks page, a red message appears near the top: “Failed to fetch bookmarks.” 5. UC.08.EX.05 – No bookmarks yet. The Bookmarks page shows a grey empty-state line: “No bookmarks found.”
Includes:	NIL
Special Requirements:	1. Bookmarks list appears within $\leq 1.0s$ after the page loads on typical broadband. 2. Remove action completes and visually updates the list within $\leq 500ms$ 3. Card layout (street, flat type, price, month) is consistent across Bookmarks and Listings detail links.
Assumptions:	1. Reasonable cap to the number of shortlists (e.g. 30) 2. The User is generally signed in when using Bookmarks
Notes and Issues:	NIL

UC.10 View Unit Details

Use Case ID:	UC.10		
Use Case Name:	View Unit Details		
Created By:	Kenny Lim	Last Updated By:	Kenny Lim
Date Created:	22/08/2025	Date Last Updated:	22/08/2025

Actor:	Logged-in User
Description:	After the User clicks on a particular unit in UC.06 (View HDB Listings), a new page is opened that shows full flat details (block, street, lease details, price, floor area, storey, flat type, etc.), affordability information, local amenities (on a map)
Preconditions:	1. User selected a unit from the listing page (UC.06) or from their Bookmarks (UC.09) 2. Detailed attributes of the unit are retrievable
Postconditions:	1. User is able to view all the various attributes of the unit
Priority:	High
Frequency of Use:	High – Estimated 5 to 10 times per user per hour

Flow of Events:	1. User clicks on a unit from the listings page 2. The System shows a loading panel while fetching details. 3. The System displays the unit title (Town, Flat Type) and a prominent price. 4. The System shows a grid of attributes (Block, Street, Storey, Floor Area, Model, Lease Commence, Remaining Lease, Month). 5. System shows affordability information (UC.12) 6. System shows local amenities (UC.11)
Alternative Flows:	NIL
Exceptions:	NIL
Includes:	View Local Amenities (UC.11) View Affordability (UC.12)
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

UC.11 View Local Amenities

Use Case ID:	UC.11		
Use Case Name:	View Local Amenities		
Created by:	Wang Cheng	Last Updated by:	Wang Cheng
Date Created:	25/08/2025	Date Last Updated:	25/08/2025
Actor:	Logged-in User		
Description:	This allows the User to view local amenities on a map, including: 1. Transport (MRT & Bus) 2. Schools 3. Clinics / Polyclinics 4. Supermarkets 5. Hawker Centres 6. Parks 7. Libraries 8. Community Clubs		
Preconditions:	The User is currently on the page where they view unit details		
Postconditions:	The User can view local amenities on a map		
Priority:	High		

Frequency of Use:	High, estimated 5-10 times per user per hour
Flow of Events:	1. The User selects a type of amenity they want to see on a map 2. Assuming the User chose “Transport” 3. MRT exits and bus stops within 1 km range of the HDB unit will be displayed on the map 4. The User scrolls the mouse wheel up 5. The System zooms in on the map
Alternative Flows:	UC.11.AF-S3: If there are no MRT exits or bus stops within 1 km range - The System displays “No Transport found within 1 km of this flat”
Exceptions:	NIL
Includes:	NIL
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

UC.12 View Affordability

Use Case ID:	UC.12		
Use Case Name:	View Affordability		
Created by:	Wang Cheng	Last Updated by:	Kenny Lim
Date Created:	25/08/2025	Date Last Updated:	19/10/2025
Actor:	Logged-in User		
Description:	This allows the User to view affordability of the HDB unit		
Preconditions:	1. The User is currently on the page where they view unit details		
Postconditions:	1. The User can view affordability successfully		
Priority:	High		
Frequency of Use:	High, estimated 5-10 times per user per hour		

Flow of Events:	1. The User is currently on unit details page 2. The User sees the section on Affordability 3. The System displays the information on the affordability of this unit, including: - Down payment required - Down payment budget - Can afford down payment - Tenure - Monthly repayment - 30% income limit - Can afford monthly - Affordability score
Alternative Flows:	NIL
Exceptions:	NIL
Includes:	NIL
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

UC.13 Use HDB Finder

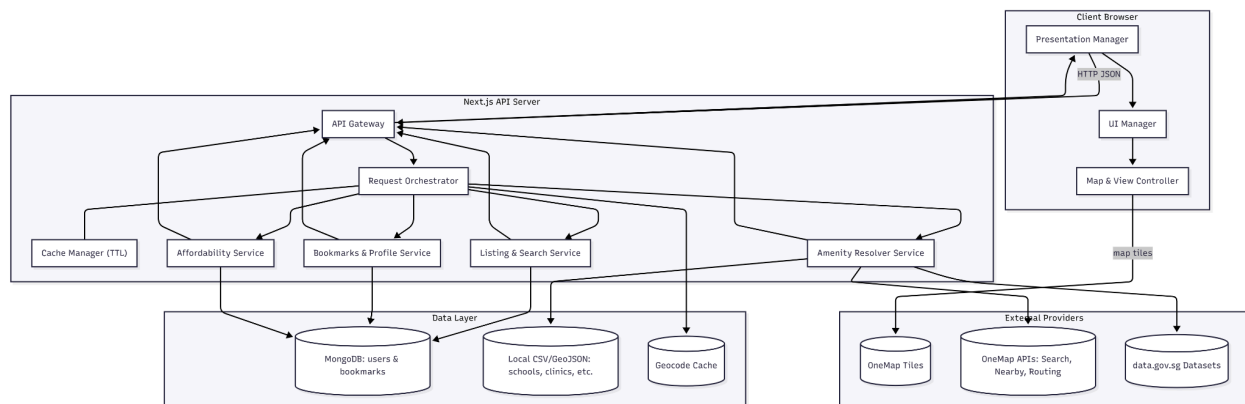
Use Case ID:	UC.13		
Use Case Name:	Use HDB Finder		
Created by:	Wang Cheng	Last Updated by:	Wang Cheng
Date Created:	25/08/2025	Date Last Updated:	9/11/2025
Actor:	Logged-in User or Guest		
Description:	The website recommends the best units for the users based on the user's preference.		
Preconditions:	1. The User is currently on the home page		
Postconditions:	1. The System displays a list of the best units based on the User's preference		
Priority:	High		
Frequency of Use:	High, estimated 5-10 times per user per hour		

Flow of Events:	1. The User is currently on the home page. 2. The User clicks “Go to HDB Finder!” button. 3. The System redirects the User to the HDB Finder page. 4. The User types the town they want into the search bar. 5. The User clicks “Add” 6. The User selects the number of rooms they want 7. The User drags the slider to set the priorities for the four variables, including: - Distance to MRT - Distance to Schools - Distance to Medical - Affordability 8. The User clicks “Find Flats” 9. The System displays the list of best units based on the User’s preferences
Alternative Flows:	NIL
Exceptions:	UC.13.EX.01: If the User have added 3 towns, and try to add another town, they will not succeed, because the User is only allowed to choose up to 3 towns.
Includes:	NIL
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

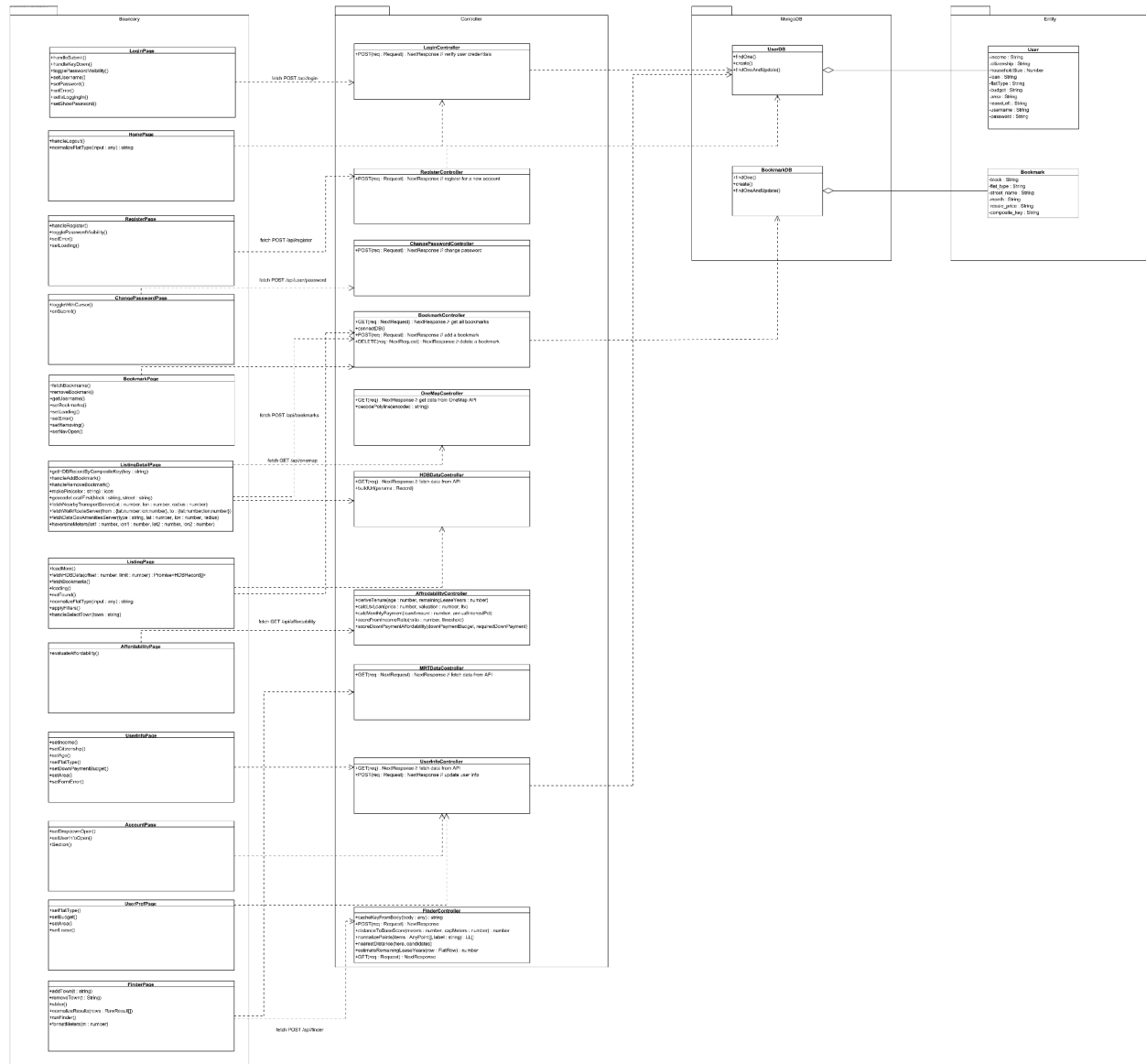
Appendix D0: Architecture Diagram

This architecture splits the app into four parts: Client, API Server, Data, and External providers, with clear responsibilities and simple, predictable flows. On the Client, the Presentation Manager renders pages and results; the UI Manager holds view state and triggers requests; and the Map & View Controller draws the interactive map. The browser fetches OneMap tiles directly for smooth pan/zoom, while data (listings, amenities, routes, bookmarks) is fetched from our APIs as JSON.

All requests enter the API Server through the API Gateway and are routed by a small Request Orchestrator to focused services: Listing & Search (resale data and detail views), Amenity Resolver (normalizes schools/clinics/MRT etc.), Affordability (scores based on user profile and price), and Bookmarks & Profile. A short-TTL Cache Manager sits beside the orchestrator to avoid recomputing heavy queries. The Data layer contains MongoDB (users, bookmarks), local CSV/GeoJSON (curated public datasets for stable, fast lookups), and a geocode cache. External providers supply base map tiles and live signals: OneMap for search/nearby/routing and data.gov.sg for official datasets. End-to-end: user action → API Gateway → Orchestrator → services (using cache, local data first, external APIs when needed) → normalized JSON back to the UI, while the map keeps rendering tiles. The result is a modular, low-latency system: UI is thin and reactive, controllers are small and stateless, and domain logic lives in one place with reliable data and clear fallbacks.



Appendix D1: UML Class Diagram



Appendix E: Test Case

Black Box Testing Approach

This approach will:

- Verify system functionality without knowledge of internal implementation
- Focus on inputs, outputs, and external behavior as specified in this SRS
- Ensure the system meets all functional requirements from the user's perspective

Key Principles:

- Testing will be based solely on requirements documented in this specification.
- Test cases will be derived from use cases, functional requirements, and business rules
- No access to source code, database schemas, or internal architecture is required for test execution

Test Case 1: User register account and log in with valid and invalid credentials

Description: User shall be able to register for an account and log into their account with valid and invalid credentials

Test Description	Expected Result	Test Result
User registers for an account	Account successfully registered	Account successfully registered
User logs in with valid credentials	User successfully logged in	User successfully logged in
User logs in with invalid credentials	Login fails with error message “Invalid credentials”	Login fails with error message “Invalid credentials”
User attempts to log in without filling in the username, password, or both	System prompts the user to fill in the missing fields before proceeding	System prompts the user to fill in the missing fields before proceeding

Test Case 2: User information, change password and log out

Description: User shall be able to update their profile information and preference in HDB Listings, change password for their account and log out.

Test Description	Expected Result	Test Result
User updates profile information	User can update their User information and preferences, and successfully save the changes	User can update their User information and preferences, and successfully save the changes
User views profile information	User can see their account username, email, and saved information and preferences	User can see their account username, email, and saved information and preferences
User changes account password	User can change their password by entering the current password and a new password	User can change their password by entering the current password and a new password
User logs out	User is successfully logged out and	User is successfully logged out and

	redirected to the frontpage	redirected to the frontpage
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Test Case 3: Homepage

Description: The system shall display all available functions on the homepage that assist the user in filtering and finding their desired HDB listings.

Test Description	Expected Result	Test Result
User click on all the available functions: View all Flats View Bookmarked Flats Go to HDBFinder! Featured neighbourhood listings Account User Information Log out	Each click successfully routes user to the appropriate section/page	Each click successfully routes user to the appropriate section/page

Test Case 4: Bookmark Listings

Description: User shall be able to save desirable listings for future viewing and comparison in the Bookmarked Listings section, as well as delete them when no longer needed.

Test Description	Expected Result	Test Result
User (Logged in) can add a listing to Bookmarked Listing	Listing is successfully saved and appears in the user's Bookmarked Listings	Listing is successfully saved and appears in the user's Bookmarked Listings
User (Logged in) can view Bookmarked Listing	All of user's saved listings are displayed	All of user's saved listings are displayed
User (Logged in) can remove a listing from Bookmarked Listing	Listing is removed and no longer appears in user's Bookmarked Listing	Listing is removed and no longer appears in user's Bookmarked Listing
User (Guest) access any of the bookmark functions	System deny access, prompts user to log in	System deny access, prompts user to log in

Test Case 5: Featured Listings and Listings Overview

Description: The system shall display featured HDB listings based on their preferences in the homepage. Each listing will display brief details, and the user can view more information by clicking on the listing.

Test Description	Expected Result	Test Result
System shall feature HDB Listings based on User's preference	Featured HDB listings are displayed to User based on their preference	Featured HDB listings are displayed to User based on their preference
HDB Listing overview will display the following to the user: neighbourhood name, flat type, cost price and affordability rating	User can see the brief details of each HDB listing	User can see the brief details of each HDB listing

Test Case 6: HDB Listing detailed information with HDB Amenities Display

Description: The system shall display the full details of the selected HDB listing, including nearby amenities. The user shall also be able to view these amenities displayed on a map.

Test Description	Expected Result	Test Result
User clicked on a desirable listing to view the full details of it.	The system shall display the following attributes for each HDB listing: neighbourhood, flat type, cost price, block, street, storey, floor area, model, lease commencement date, remaining lease, and listing posted month. The system displays all listed attributes accurately with corresponding values, and loads a map showing nearby amenities within a 1 km radius.	The system displayed all information and the nearby amenities on a map to the user.
User filtered the HDB amenities display to view specific categories such as transport, schools, clinics/polyclinics, supermarkets, hawker centres, parks, libraries, or community clubs nearby.	The system shall display the selected types of amenities around the HDB unit. The user shall be able to explore the area by clicking and dragging on the map.	The system shall display the selected types of amenities around the HDB unit. The user shall be able to explore the area by clicking and dragging on the map.

Test Case 7: View all Flats filtering

Description: The system shall allow the user to filter HDB listings by the following attributes, which may be used individually or in combination: neighbourhood, price range, and flat type. The system shall display the HDB Listings based on the user's filtering.

Test Description	Expected Result	Test Result
Add "Bishan" to Town Set "Price range" to \$300 000 - \$350 000 Set "Number of rooms" to 3 ROOM Click "Find Flats"	HDB units are displayed accurately according to the selected filters.	More than 20 HDB units are displayed according to the applied filters.
Click "Clear town"	The displayed listings are reset, showing the initial flats with no filters applied..	More than 20 HDB units are displayed with varying towns, flat types, and prices.
Set "Price range" to \$300 000 - \$350 000 Click "Find Flats"	HDB units are displayed accurately according to the selected filters.	More than 20 HDB units are displayed, including a mix of 3-room and 4-room flats from different towns.
Add "Bukit Merah" to Town Click "Find Flats"	HDB units are displayed accurately according to the selected filters.	More than 20 HDB units are displayed with a variety of flat types and prices.
Set "Number of rooms" to 3 ROOM Click "Find Flats"	HDB units are displayed accurately according to the selected filters.	More than 20 HDB units are displayed with a mix of towns and prices.

Test Case 8: HDBFinder filtering

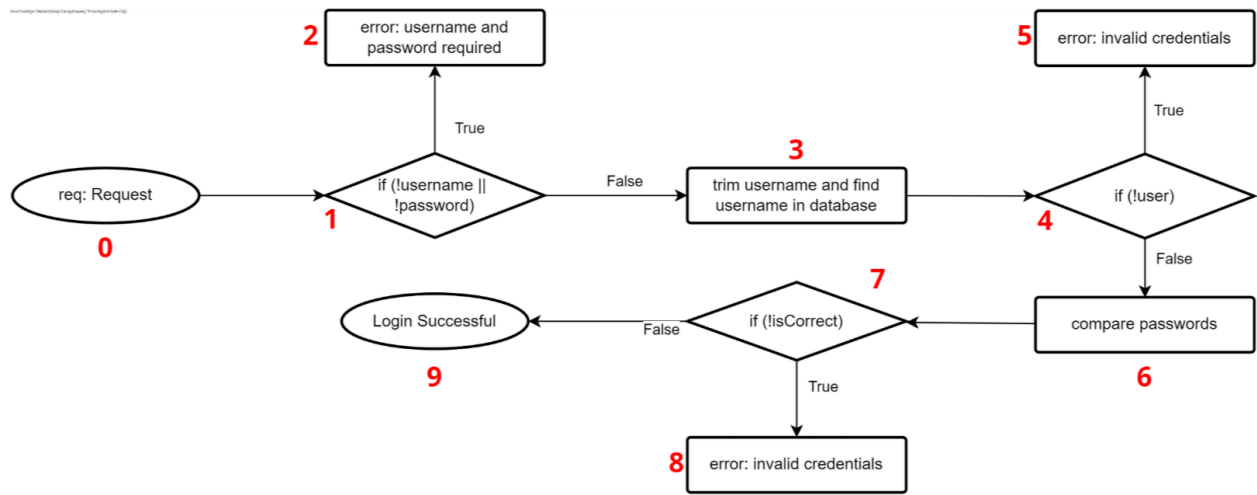
Description: The system shall allow the user to assign a priority rating (on a scale of 1 to 10) to the following attributes: distance to MRT, distance to schools, distance to medical facilities, and affordability rating. The system will display the HDB Listings based on the user's list of priority for amenities.

Test Description	Expected Result	Test Result
Add "Bishan" to town Choose 5 Room Flat Type Set "Distance to MRT" to priority 10 Set other criteria to priority 0 Click "Find Flats"	HDB units are displayed and sorted from low to high according to distance to MRT	5 Room HDB units are displayed and sorted from low to high according to distance to MRT.
Add "Bishan" to town Choose 4 Room Flat Type Set "Distance to MRT" to priority 10 Set "Distance to Schools" to priority 5 Set other criteria to priority 0 Click "Find Flats"	HDB units are displayed and sorted from low to high according to distance to MRT. Among those units with the same distance to MRT, they are sorted according to distance to schools from low to high	4 Room HDB units are displayed and sorted from low to high according to distance to MRT. Among those units with the same distance to MRT, they are sorted according to distance to schools from low to high.
Add "Bishan" to town Set Affordability to 8. Set other criteria to priority 0	All flats in Bishan that have an affordability score of 8 and above are included.	All flats in Bishan are sorted based on affordability.
Add "Bukit Timah" to town Set the 4 criteria to arbitrary values between 0 and 10 Click "Find Flats"	HDB units are displayed and sorted from high to low according to the computed score.	HDB units are displayed and sorted from high to low according to the computed score.

Add three towns Search and add another town	Cannot add any more town	Cannot add any more town
Add “Bukit Timah” Search for “Bukit Timah”	“Bukit Timah” does not appear in the drop-down list, and hence cannot be added again	“Bukit Timah” does not appear in the drop-down list, and hence cannot be added again
Click “View details”	HDB unit details are displayed	HDB unit details are displayed

White Box Testing – Basis Path Testing

Control Flow Graph of Login Process



Compute Cyclomatic Complexity

$$CC = |\text{decisionPoint}| + 1 = 3 + 1 = 4$$

Basis Paths

We conduct tests for each of the paths, so that we can achieve 100% coverage of all possible branches.

Basis Paths	Description	Test Input	Expected Result	Test Result
0, 1, 3, 4, 6, 7, 9	baseline path, login successful	Enter an existing username Enter the correct password Click “Login”	Login Successful	Login Successful
0, 1, 2	username or password is	Enter an existing username	The System displays “Please	The System

	missing	Leave password blank Click "Login"	fill out this field"	displays "Please fill out this field"
0, 1, 3, 4, 5	user does not exist	Enter a non-existing username Enter password as "11111111" Click "Login"	The System displays "invalid credentials"	The System displays "invalid credentials"
0, 1, 3, 4, 6, 7, 8	password is incorrect	Enter an existing username Enter a wrong password Click "Login"	The System displays "invalid credentials"	The System displays "invalid credentials"